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Submitted via email

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RE: Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

I. Introduction

The Working for Advanced Transmission Technologies (WATT) and Advancing Modern Powerlines (AMP) Coalitions respectfully submit these comments in response to the request for information (RFI) issued February 3, 2025, on the development of an Artificial Intelligence (AI) Action Plan. The Coalitions are focused on facilitating the adoption of advanced technologies on the US electric transmission system that improve reliability, lower cost, and accelerate integration of new load and generation resources.

II. About the WATT Coalition

The WATT Coalition advocates for policy that supports the wide deployment of Grid Enhancing Technologies (GETs) to accelerate the lower energy costs and enable economic growth. GETs are hardware and/or software that increases the capacity, efficiency, and maintains the reliability of the electric grid. Learn more at watt-transmission.org. WATT's membership includes Dynamic Line Ratings, Advanced Power Flow Control and Transmission Topology Optimization vendors, as well as independent power producers, clean energy investors and utilities.

III. About the AMP Coalition

The AMP Coalition is an ad hoc coalition of CTC Global Corporation, TS Conductor, VEIR, and MetOx. AMP's goal is to further the use of High Performance Conductors as a tool for modernizing and increasing grid capacity, as well as improving the overall resilience, reliability, and energy efficiency of the grid.

IV. The Current Generation Interconnection and Transmission Planning Processes Are Creating Unacceptable Delays for Data Center Development

Shortcomings in transmission planning and generation interconnection processes have emerged as the critical bottleneck in the AI revolution. The unprecedented energy demand necessary to train and run more powerful models is stretching the grid to its limits. Late last year, the Department of Energy released a study finding that data centers would use as much as 12% of

total US electricity demand by 2028, up from 4.4% in 2023.¹ On top of this new demand, US grid infrastructure is rapidly aging.

In PJM, the 13-state transmission region which includes major data center clusters in Ohio and Virginia, two-thirds of all electrical assets are at least forty years old, and one-third are at least 50 years old.² PJM is projected to host as much as 70% of US data center growth.³ This demand – particularly in northern Virginia’s ‘data center alley’, home to as much as 70% of all internet traffic – is driving the need for new greenfield transmission.⁴ But transmission projects are notoriously difficult to site: they are time and capital intensive, cross multiple municipalities, and often involve use of eminent domain. For example, the Piedmont Reliability Project, a planned transmission line running from southeast Pennsylvania into Maryland, has sparked backlash from landowners and politicians of both parties.⁵ New greenfield transmission projects can be contentious – a fundamental reality that can be managed, but not eliminated, by permitting reform. Consequently, utilities are quoting wait times as long as seven years for new large loads to interconnect.⁶ The US AI Action Plan should prioritize support for near-term and long-term transmission solutions in tandem to ensure grid operators can bring new load online quickly and unlock the transformative power of AI.

Behind-the-meter (BTM) generation options are not a panacea for data center operators, who have turned to a wide array of agreements with generation developers to accelerate the powering of these facilities. But, at least for the foreseeable future, nearly all such arrangements will still rely on grid interconnection. Going fully off-grid poses unique risks which many large interconnection customers will not accept. AI data centers need to operate at extremely high levels of reliability – often at least 99.999 percent (‘five nines’) uptime.⁷ That standard allows for less than seven minutes of outage in a year. Any off-grid generation solution would require maintenance outages exceeding that limit and offer less redundancy compared to the grid. Fully BTM solutions would also introduce project risks and regulatory uncertainty.⁸ For the vast majority of operators, grid interconnection with generator backup is preferable to fully off-grid data centers.

¹ Lawrence Berkeley National Laboratory, “2024 United States Data Center Energy Usage Report,” December 2024, <https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf> at 5.

² PJM, “*The Benefits of the PJM Transmission System*,” at 5, April 16, 2019.

³ Boston Consulting Group, “*Breaking Barriers to Data Center Growth*,” January 20, 2025, <https://www.bcg.com/publications/2025/breaking-barriers-data-center-growth>.

⁴ Prince William Times, “*Opposition mounts to ‘spiderweb’ of new transmission lines needed to power data center alley*,” October 25, 2023, https://www.princewilliamtimes.com/news/opposition-mounts-to-spiderweb-of-new-transmission-lines-needed-to-power-data-center-alley/article_b7da863c-dfac-516d-9dfe-9adc34b8c8bd.html.

⁵ WBAL, “*PSEG files application with PSC seeking authority to build Maryland Piedmont Reliability Project*,” December 31, 2024, <https://www.wbaltv.com/article/piedmont-powerline-project-pseg-files-application-psc/63313157>.

⁶ Bloomberg, “*Data Centers Face Seven-Year Wait for Dominion Power Hookups*,” August 29, 2024, <https://www.bloomberg.com/news/articles/2024-08-29/data-centers-face-seven-year-wait-for-power-hookups-in-virginia?embedded-checkout=true>

⁷ SemiAnalysis, “*Datacenter Anatomy Part 1: Electrical Systems*,” October 24, 2024, <https://semianalysis.com/2024/10/14/datacenter-anatomy-part-1-electrical/>.

⁸ Id. 3,

“Speed-to-power” has emerged as the key priority for data center developers – outpacing considerations like access to land, water, chips, or fiber, and the price of electricity.⁹ But the US electricity system is struggling to match the pace of AI expansion. Planning, siting and permitting new transmission lines requires, on average, seven to ten years.¹⁰ In some extreme examples, projects have taken up to seventeen years to complete due to permitting delays and other challenges.¹¹ There is an inherent mismatch between the timelines required by AI use and the pace of growth of our electricity system. Making it easier to permit and build large transmission and generation projects is a crucial part of the solution – permitting reform is needed. But the administration should also aim to maximize the generation already connected to the grid by taking advantage of advanced transmission technologies (ATTs).

V. Advanced Transmission Technologies (ATTs) Can Offer Some Immediate Relief to Data Center and Power Generation Developers

ATTs can maximize the useful capacity of existing grids and infrastructure. These technologies are lower-cost and faster to deploy than either new transmission lines or new generation resources.¹² ATTs include Grid Enhancing Technologies (GETs) and High Performance Conductors (HPCs).

a. About Grid Enhancing Technologies

Grid Enhancing Technologies, or GETs, are hardware and software that increase the capacity, efficiency and reliability of the transmission system. These technologies are installed on existing transmission infrastructure to give operators more situational awareness and control over the grid.

- **Power flow control technologies** actively balance the flow on transmission lines. The hardware can intelligently raise or lower the impedance, or the opposition to current, in real time to ensure that power is delivered on lines that have the capacity to carry it. **Advanced power flow control (APFC)** expands on this function with enhancements such as faster and more flexible deployment options, easy scaling to meet the size of the need, and ability to relocate when needed elsewhere on the grid.
- **Dynamic Line Rating (DLR)** monitors ambient conditions which heat or cool transmission lines to calculate the true capacity of transmission lines, based on their

⁹ McGeady, Cy et al., “*The Electricity Supply Bottleneck on US AI Dominance*,” March 2025, https://csis-website-prod.s3.amazonaws.com/s3fs-public/2025-03/250303_McGeady_Electricity_Bottleneck.pdf?VersionId=LsCkrctQBekLSYFPxus3XWdmTCnmIr9Z

¹⁰ Lisa Kelso, Testimony before the Public Utilities Commission of Ohio, “*Direct Testimony of Lisa Kelso on Behalf of Ohio Power Company*,” Case No. 24-508-EL-ATA, filed May 13, 2024, <https://dis.puc.state.oh.us/ViewImage.aspx?CMID=A1001001A24E13B43501B00952>.

¹¹ Wall Street Journal, “*The U.S. ‘Fast-Tracker’ a Power Project. After 17 Years, It Just Got Approved.*” May 18, 2023, <https://www.wsj.com/articles/the-u-s-fast-tracked-a-power-project-after-17-years-its-nearing-approval-1a7edb86>

¹² The Working for Advanced Transmission Technologies (WATT) Coalition and The Advancing Modern Powerlines (AMP) Coalition, *Benefits of ATTs Compared to Traditional Upgrades*, August 2024, [WATT-and-AMP-Benefits-of-ATTs-compared-to-traditional-upgrades.pdf](#), at 1

thermal limits. In windy regions where transmission capacity is critical for wind power development, DLR often allows significantly more power flow than a static rating over the course of the year but also detects when flows should be reduced to continue safe and reliable operation in extreme heat or other conditions.

- **Transmission topology optimization (TTO)** software models the grid's network and power flow conditions to identify ways to reroute power flow around congested or overloaded transmission elements. These "reconfigurations" are implemented by switching on or off existing high-voltage circuit breakers. By more evenly distributing flow over the network, topology control increases the transfer capacity of the grid, like Google maps redirecting cars in traffic.

b. About High Performance Conductors

High Performance Conductors (HPCs) is a term that encompasses modern conductor technologies which have greater performance characteristics when compared to traditional conductors, including increased capacity, higher efficiency, and less thermal sag.

- **Carbon and Composite Core Conductors**, also known as Advanced Conductors, are overhead, bare conductors that use a trapezoid shaped wire of annealed aluminum to carry electrical current and use a carbon or composite core for support. Used commercially for 20 years, the conductors are deployed in over 60 countries across 5 continents. Carbon and composite core conductors have three key advantages over traditional conductors: stronger and lighter weight cores, which allows for more aluminum to be added to the conductor, doubling the capacity; 20% or greater efficiency, and half as much thermal sag.
- **Superconducting transmission lines** use a class of metallic compounds that exhibit negligible resistance when cooled using liquid nitrogen, enabling very low losses and very high power-flow capacities. Developed in the 1980s, countries in Europe and Asia, as well as the U.S., have installed superconductors over the past 20 years. Superconductors have three key advantages over traditional conductors: increase capacity 5-10 times at lower voltages, reducing substation build and cost, no thermal sag and line sag does not vary with ambient weather conditions or exposure to elements, and they are at least 50% more energy efficient.

These technologies are low-hanging fruit: they are cost-effective and can help bring new load and new generation online quickly by better utilizing the grid or increasing the power density of existing corridors. Not only will they help meet the electricity demand from new data centers, but they will also be a crucial tool as data center developers move to larger compute clusters.

GETs and HPCs have well-established benefits but are not yet widely deployed in the US. Reconductoring with carbon or composite core conductors can increase the capacity of an

existing line by 50-110% at a cost less than half that of greenfield transmission and rebuilding with modern superconductors could increase the capacity of lines by over 500%.¹³ Deploying GETs including dynamic line rating (DLR) created an average increase of 47% in line capacity in Massachusetts, and up to 60% increase in line capacity for the New York Power Authority.¹⁴ DLR is currently the subject of an advanced notice of proposed rulemaking at FERC.¹⁵

These technologies are widely deployed in Europe and Asia, but uptake has been slower in the US. Barriers to deployment include a lack of familiarity from grid planners, regulators, and utilities.

HPCs can offer an excellent solution for utilities dealing with significant load growth. In 2015, rapid population growth drove record power demand in the Rio Grande Valley. The area's sole source of power was a pair of legacy transmission lines – each over thirty years old. AEP was able to replace 240 line-miles with advanced conductors, nearly doubling the previous power capacity. The project was completed millions of dollars under budget, eight months ahead of schedule, and without any interruptions for consumers.¹⁶

As interconnection requests have increased over the past decade, generation developers have experienced longer waits, higher interconnection costs, and more frequent cancellations stemming in part from limited transmission headroom.¹⁷ According to the Lawrence Berkeley National Lab, over two terawatts of generating resources are currently languishing in interconnection queues.¹⁸ A Brattle Group study of states in the Southwest Power Pool found that uptake of GETs could allow double the generating resources to come online relative to a base case.¹⁹ GETs would also enable operators to reduce curtailment of new and existing generation. Similarly, using HPCs to replace just 25% of the oldest infrastructure on the US transmission system would allow the interconnection of an additional 27 GW of generation annually over the next decade.²⁰

¹³ GridLab, “*The 2035 Report: Reconductoring with advanced conductors can accelerate the rapid transmission expansion required for a clean grid*,” April 2024, www.2035report.com/wp-content/uploads/2024/04/GridLab_2035-Reconductoring-Technical-Report.pdf at 17.

¹⁴ The Working for Advanced Transmission Technologies (WATT) Coalition and The Advancing Modern Powerlines (AMP) Coalition, *Benefits of ATTs Compared to Traditional Upgrades*, August 2024, WATT-and-AMP-Benefits-of-ATTs-compared-to-traditional-upgrades.pdf, at 2

¹⁵ Fact Sheet | ANOPR: Implementation of Dynamic Line Ratings, June 27, 2024, <https://www.ferc.gov/news-events/news/ferc-seeks-comment-potential-dlr-framework-improve-grid-operations-fact-sheet#FS>.

¹⁶ Power Engineers, “*Lower Rio Grande Valley 345 kV Reconductor Project*,” (n.d.).

¹⁷ Gorman, W. et al. (2025) ‘Grid connection barriers to renewable energy deployment in the United States’, *Joule*, 9(2). doi:10.1016/j.joule.2024.11.008.

¹⁸ Rand, Joseph et al., “*Queued Up: 2024 Edition Characteristics of Power Plants Seeking Transmission Interconnection As of the End of 2023*,” April 2024, https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_R2.pdf.

¹⁹ Tschuida, T. Bruce et al., “*Unlocking the Queue with Grid-Enhancing Technologies*,” February 1, 2021, https://watt-transmission.org/wp-content/uploads/2021/02/Brattle___Unlocking-the-Queue-with-Grid-Enhancing-Technologies_Final-Report_Public-Version.pdf#90.pdf.

²⁰ Jay Caspary and Jesse Schneider, *Grid Strategies, Advanced Conductors on Existing Transmission Corridors to Accelerate Low Cost Decarbonization*, ACORE, March 2022, https://acore.org/wp-content/uploads/2022/03/Advanced_Conductors_to_Accelerate_Grid_Decarbonization.pdf at 19-20.

States as politically and geographically diverse as Montana, Minnesota, Virginia, and California have approved measures to expand the use of GETs and HPCs due to their cost saving benefits.²¹ FERC has also stressed the need for grid operators and utilities to consider GETs and HPCs. In Order No. 2023, the agency's landmark interconnection reform, transmission providers are required to evaluate HPCs and a select number of GETs technologies in assessing grid upgrades needed for generator interconnection.²² In Order No. 1920-A, the transmission planning rule passed last May and unanimously reaffirmed in November, transmission providers were required to assess a range of GETs and HPCs in their planning processes.²³

The speed of the AI transition demands that, in addition to installing vast new sources of generation, the US maximizes the potential of the existing grid and prudently expand its capacity in new areas as needed. Interconnection queues, siting and permitting challenges, and supply chain constraints around critical technologies, including large power transformers, have expanded the lead times to build all forms of new generation and grid components.²⁴ ²⁵ HPCs and GETs offer near-term transmission capacity expansion solutions for the existing grid and new projects, with installation times of can 1-3 years and six months, respectively.²⁶

In recognition of the important role ATTs can play in supporting the development of grid infrastructure and capacity needed to support AI development, the AI Action Plan should include the following recommendations:

- **DOE should continue and expand its support of HPC and GETs projects.** As of April 2024, DOE's Grid Resilience and Innovation Partnerships program supported over \$2 billion worth of GETs and HPC projects.²⁷ Similarly, the Loan Programs Office has supported \$23 billion in transmission projects, including reconductoring, under the Title 17 Energy Infrastructure Reinvestment program.²⁸ Funds in other existing programs, including the Innovative Energy Loan Program and the Tribal Energy Finance Program could be dedicated to GETs or HPCs.²⁹ Federal support for these technologies, whether

²¹ Pew Charitable Trusts, "With U.S. Energy Grid Under Strain, Governments Promote Technology Solutions," August 2, 2024, <https://www.pewtrusts.org/en/research-and-analysis/articles/2024/08/02/with-us-energy-grid-under-strain-governments-promote-technology-solutions>.

²² Order No. 2023, Docket No. RM22-14, at 1011.

²³ Order No. 1920-A, Docket No. 21-17, at 506.

²⁴ Resources for the Future, "Modeling Obstacles to Energy Infrastructure for Improved Policy Analysis," September 2024, https://media.rff.org/documents/Report_24-16_-_Updated_Draft.pdf.

²⁵ Department of Energy "Pathways to Commercial Liftoff: Innovative Grid Deployment," April 2024, https://liftoff.energy.gov/wp-content/uploads/2024/04/Liftoff_Innovative-Grid-Deployment_Final_4.15.pdf.

²⁶ Department of Energy "Pathways to Commercial Liftoff: Innovative Grid Deployment," April 2024, https://liftoff.energy.gov/wp-content/uploads/2024/04/Liftoff_Innovative-Grid-Deployment_Final_4.15.pdf.

²⁷ Department of Energy "Pathways to Commercial Liftoff: Innovative Grid Deployment," April 2024, https://liftoff.energy.gov/wp-content/uploads/2024/04/Liftoff_Innovative-Grid-Deployment_Final_4.15.pdf.

²⁸ Department of Energy, "Deal Digest: LPO's Latest Conditional Commitments Help Keep Power Affordable Through New Generation and Expanded Transmission for Utility Customers in Twelve States," January 16, 2025, <https://www.energy.gov/lpo/articles/deal-digest-lpos-latest-conditional-commitments-help-keep-power-affordable-through-new>.

²⁹ Watt Coalition, "Federal Funding for Advanced Transmission Technologies," accessed February 28, 2025, <https://watt-transmission.org/wp-content/uploads/2024/08/WATT-and-AMP-Federal-Funding-for-Advanced-Transmission-Technologies.pdf>.

through grants or loans, will help accelerate their urgently needed deployment, and improve utility comfort with the technology along with providing lessons learned for future deployments.

- **DOE should consider implementing a conductor efficiency and/or a resistance-based standard.** Congress has already vested DOE with the authority to establish energy-efficiency standards for certain electrical equipment. Standards have been developed for over 70 product categories, including boilers, central air conditioners, and distribution transformers.³⁰ Establishing efficiency or resistance-based standards for transmission would provide an important signal for utilities and manufacturers.
- **The administration should leverage the authority of the national laboratories.** Skepticism from industry planners and utility regulators is a key barrier to deployment of GETs and HPCs. The federal government should leverage institutions, such as the national labs, to lead on bringing these important tools into the mainstream. For example, they should be considered in assessments of technology readiness such as the NREL's Annual Technology Baseline.³¹
- **The administration should support the future transmission workforce.** The administration should examine whether our workforce is equipped for the widespread manufacture and deployment of these advanced technologies. Congress and the Departments of Education and Labor should invest in workforce training programs for power system engineers, electricians, and linemen.
- **The administration should require other federal agencies to adopt DOE's categorical exclusion for reconductoring and rebuilds.** DOE established a NEPA Categorical Exclusion to streamline the permitting process for projects reconductoring or rebuilding transmission lines, increasing capacity much more quickly by utilizing existing rights-of-way and potentially existing infrastructure, such as transmission towers. DOE has limited control over federal lands and other agencies such as the Department of Interior should adopt similar categorical exclusions.
- **FERC should adopt a shared savings incentive.** As previously discussed, utilities' well-known capex bias can discourage investment in certain ATTs. A shared savings incentive rewards utilities for saving ratepayers money. Utilities would receive a portion of the savings derived from use of ATTs as determined through a standard cost-benefit analysis. This incentive could save consumers billions of dollars per year.³²

³⁰ Department of Energy, "Standards and Test Procedures," <https://www.energy.gov/eere/buildings/standards-and-test-procedures>.

³¹ GridLab, "Supporting Advanced Conductor Deployment: Barriers and Policy Solutions," April 9, 2024, <https://www.2035report.com/wp-content/uploads/2024/05/5.3-Reconductoring-policy-report.pdf>.

³² Gramlich, Rob et al., "A Roadmap for Advanced Transmission Technology Adoption," September 2024, <https://ceepr.mit.edu/wp-content/uploads/2024/09/MIT-CEEPR-RC-2024-06.pdf>.

Respectfully,

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